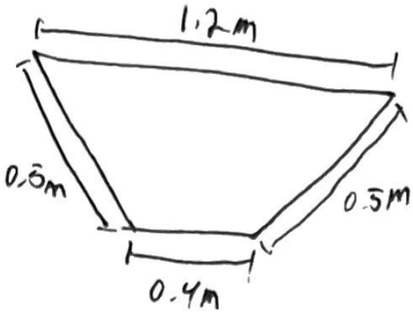
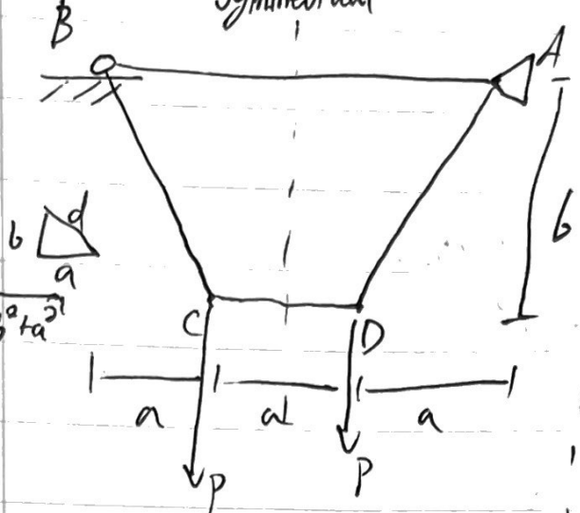


A2

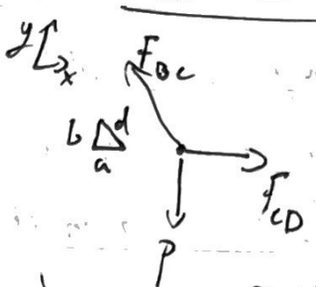
Symmetrical



$a = 0.4m$ $b = 0.3m$

$P = 25kN$

Joint C

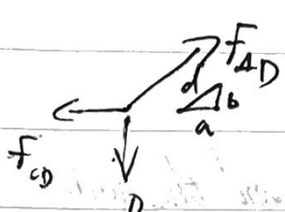


$\Sigma F_y = 0$ $\Sigma F_x = 0$

$-P + \frac{b}{d}F_{BC} = 0$ $F_{CD} - \frac{a}{d}F_{BC} = 0$

$F_{BC} = \frac{Pd}{b}$ T $F_{CD} = \frac{a}{d}F_{BC}$ T

Joint D



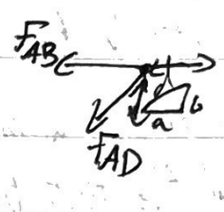
$F_{AD} = F_{BC}$

$\Sigma F_y = 0$

$-P + \frac{b}{d}F_{AD} = 0$

$F_{AD} = \frac{Pd}{b}$ T

Joint A



$\Sigma F_x = 0$ $\Sigma F_y = 0$

~~$F_{AB} - \frac{a}{d}F_{AD} = 0$~~

~~$F_{AB} = \frac{a}{d}F_{AD}$ C~~

~~Joint B~~

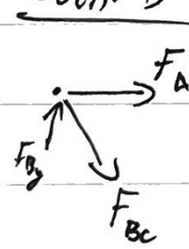
$-F_{AB} + F_{A_x} - \frac{a}{d}F_{AD} = 0$

$F_{AB} = F_{A_x} - \frac{a}{d}F_{AD}$ $F_{AB} = -\frac{a}{d}F_{AD}$ C

$-\frac{b}{d}F_{AD} + F_{A_y} = 0$

$F_{A_y} = \frac{b}{d}F_{AD}$

Joint B



$F_{AB} = -\frac{a}{d}F_{AD} = -\frac{0.4m}{0.5m} \cdot 41.667kN = \boxed{33.33kN = F_{AB}}$

$F_{AD} = \frac{Pd}{b} = \frac{25kN \cdot 0.5m}{0.3m} = \boxed{41.667kN = F_{AD}}$

$F_{BC} = F_{AD} = \boxed{41.667kN = F_{BC}}$ — Largest

$F_{CD} = \frac{a}{d}F_{BC} = \frac{0.4m}{0.5m} \cdot 41.667kN = \boxed{33.33kN = F_{CD}}$